

HISTOLOGY

Characteristic histologic features of chancroid include three vertically arranged zones:

- A superficial necrotic zone,
- A middle zone of new blood vessel formation, and
- A deep zone composed of a dense infiltrate of lymphocytes and plasma cells.

Tissue biopsy is not recommended for routine diagnosis, but histologic examination may help exclude malignancy in non-healing or atypical ulcers. Staining with *H. ducreyi*-specific monoclonal antibodies can demonstrate the organisms primarily within the granulocytic infiltrate and fibrin of the ulcer.

DIFFERENTIAL DIAGNOSIS (Box 202-1)

The three classic causes of genital ulceration are:

- *Haemophilus ducreyi* (chancroid),
- *Treponema pallidum* (syphilis), and
- Herpes simplex virus (HSV types 1 and 2).

The clinical presentation of these infections can vary widely in both men and women, making clinical diagnosis accurate in only a minority of cases. The etiology of genital ulcers also varies significantly by geographic region. In industrialized countries, isolated painful genital ulcers are most commonly caused by herpes simplex virus. In many cases, no pathogen is identified, but co-infections—such as *H. ducreyi* with *T. pallidum* (ulcus mixtum) or HSV—are not uncommon.

COURSE AND PROGNOSIS

Chancroid is typically self-limited, and systemic dissemination does not occur. However, without treatment, genital ulcers and inguinal abscesses may persist for years. Local pain is the most common symptom. If there is no clinical improvement within one week of starting therapy, possible explanations include:

- Incorrect diagnosis,

- Co-infection with another pathogen,
- Concomitant HIV infection,
- Poor adherence to treatment, or
- Infection with a resistant strain of *H. ducreyi*.

Infection does not confer lasting immunity, and reinfection is possible. To prevent reinfection, patients should be counseled on proper condom use.

COMPLICATIONS

Approximately half of untreated patients experience spontaneous resolution without complications. However, delayed treatment may lead to various complications (see Table 202-2):

TABLE 202-2 Complications of Chancroid

- **Painful inguinal adenitis** (up to 50% of cases)
- **Spontaneous rupture of inguinal buboes**, leading to large abscesses and fistula formation (rare)
- **Spread of *Haemophilus ducreyi* to distant sites** via autoinoculation, causing kissing ulcers or extragenital lesions (in 50% of male patients)
- **Acute conjunctivitis** (very rare)
- **Bacterial superinfection** (including anaerobes), potentially causing extensive tissue destruction (rare)
- **Scarring leading to phimosis** (rare)
- **Erythema nodosum** (very rare)
- **Enhanced transmission of HIV** (3-fold to 10-fold increased risk)

DIAGNOSTIC APPROACH

Serologic testing for *H. ducreyi* currently has limited utility in routine clinical diagnosis but may be valuable in epidemiologic studies to detect past infection.

Polymerase chain reaction (PCR) assays demonstrate higher sensitivity than bacterial culture for diagnosing chancroid. A multiplex PCR assay (Roche

Molecular Systems, Alameda, CA) has been developed to simultaneously detect DNA from:

- *H. ducreyi*,
- *Treponema pallidum*, and
- Herpes simplex virus types 1 and 2.

This assay shows high sensitivity (98.4%) and specificity (99.6%) for *H. ducreyi*. Although these PCR methods are not yet commercially available, they represent a promising tool for the evaluation of genital ulcers.

In resource-limited settings where diagnostic testing is unavailable, the World Health Organization recommends a syndromic management approach for patients with genital ulcer disease.